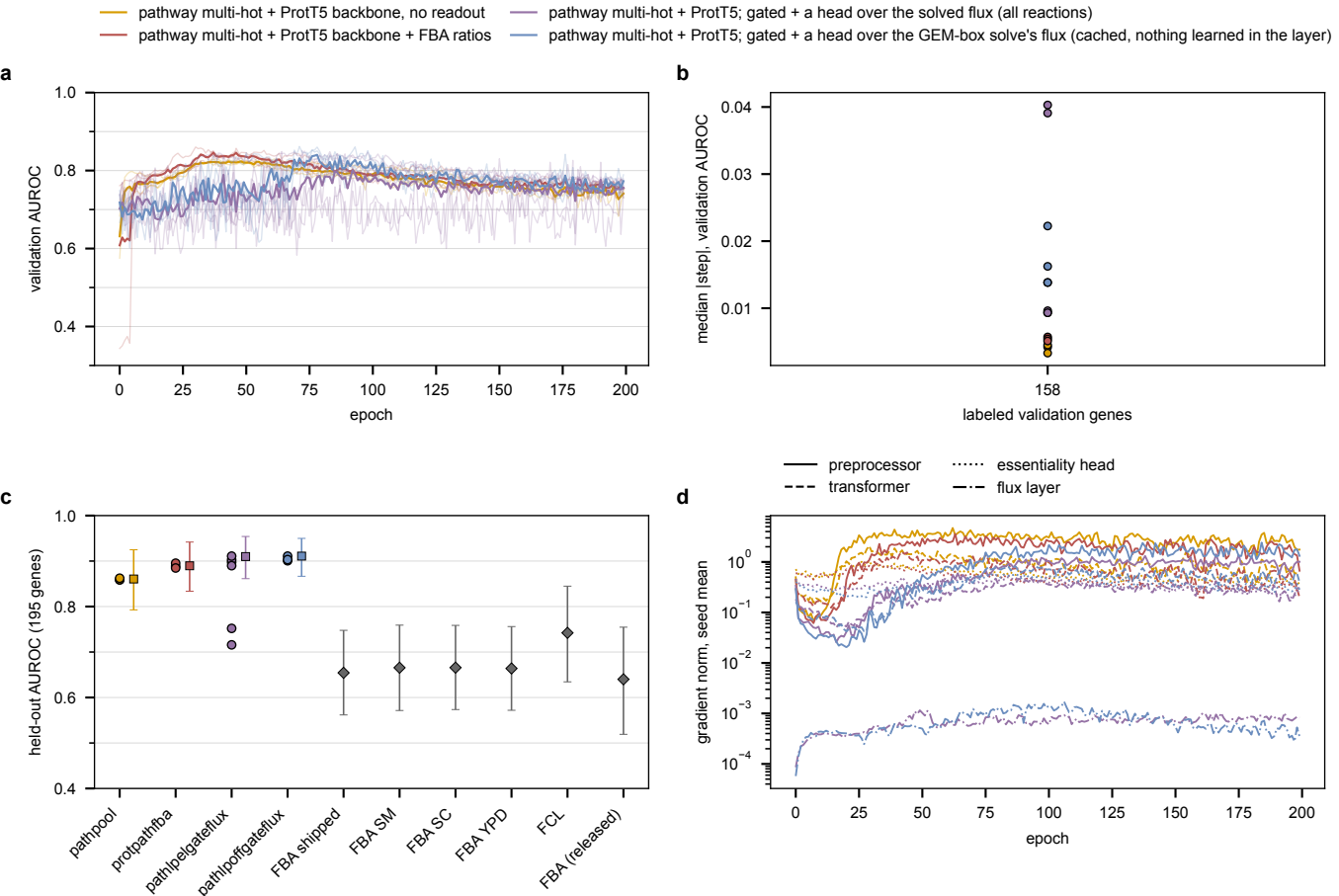

pathway multi-hot + ProtT5 backbone, no readout
pathway multi-hot + ProtT5 backbone + FBA ratios
pathway multi-hot + ProtT5; gated + a head over the solved flux (all reactions)
pathway multi-hot + ProtT5; gated + a head over the GEM-box solve's flux (cached, nothing learned in the layer)
a
validation AUROC
epoch
b
median |step|, validation AUROC
labeled validation genes
158
c
held-out AUROC (195 genes)
pathpool
protpathfba
pathlpelgateflux
pathlpoffgateflux
FBA shipped
FBA SM
FBA SC
FBA YPD
FCL
FBA (released)
d
preprocessor
transformer
essentiality head
flux layer
gradient norm, seed mean
epoch